

PAPER PRESENTATION · LEARNABILITY OF REASONING

How Far Can Transformers Reason?

The globality barrier and inductive scratchpad

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THE RESULT IN ONE SLIDE

One number predicts what a transformer can learn from scratch

THE BARRIER

$\geq n$

tokens must be read jointly before anything predicts the cycle task's label (Lemma 1)

A 10M-parameter GPT-2-style model makes no progress at $n = 7$ within 100k iterations.

THE FIX

3

tokens per step once a depth-first-search scratchpad is supervised (Lemma 3)

The same task is then learned. Unsupervised scratchpad tokens do not help (Thm 2).

THE EXTRAPOLATION

4 $\rightarrow$ 26

digits of addition, trained short and tested long, with an inductive scratchpad

Parity: trained up to 30 bits, correct at 50–55 bits.

Expressivity was never the obstacle: the cycle task is representable. The obstacle is how many tokens must be inspected at once.

Which targets can a transformer learn from scratch?

WHAT EXPRESSIVITY ANSWERS

Do good weights exist?

Constant depth with log precision computes within TC^0 ; log depth reaches TC^1 ; a polynomial-length scratchpad computes anything in P .

WHAT THIS PAPER ASKS

Will training find them?

From random initialization, by gradient descent, with polynomially many samples and steps, for one given input distribution.

THE TEST CASE

Composing syllogisms: given facts $a \Rightarrow b$, $b \Rightarrow c$, ..., does $a \Rightarrow z$? Equivalently, are two vertices connected in a graph of n edges?

Reasoning is composition, and composition is where learning stalls

What goes wrong today

Trained on answers alone, a transformer on random-graph syllogisms reaches about 80% by a degree shortcut (a heuristic alone scores about 82%) and is at chance out of distribution.

What theory offered

Expressivity bounds say what weights exist. Learning measures (SQ dimension, leap) cover function classes or fully connected nets, not one task on a transformer.

What this paper adds

A per-task measure for transformers, a proved barrier case, a typology of scratchpads by how they are supervised, and a format that extrapolates.

It turns "use chain-of-thought" into a testable condition on what each written step may depend on.

Four practical readings, each tagged by its evidence

INTERPRETATION

Answer-only data invites shortcuts

When the label needs many tokens at once, models latch onto local proxies such as vertex degree, then fail out of distribution.

THEOREM + CONJECTURE

Free thinking tokens are not enough

Scratchpad tokens trained only through the final answer do not break the barrier (Thm 2 for one task, Conj. 2 in general).

LEMMA

Write steps that read $O(1)$ tokens

A scratchpad helps when every step depends on a constant number of tokens: running parity needs 2, depth-first search needs 3.

EXPERIMENT

Make each step position-free

If each step sees only the question and the previous state, the learned step repeats at any length: 30 $\rightarrow$ 55-bit parity.

A data lever too: mixing cycle sizes 2–7 started weak learning within 30k iterations; size 7 alone showed none in 100k.

Three obstacles to a learnability theory for transformers

- 1 Existing measures describe classes, not tasks**
SQ dimension and cross-predictability score a family of targets. A transformer trained from a fixed initialization faces one distribution.
- 2 The architecture changes what is easy**
Averaging attention gives token frequencies almost for free. A measure for transformers has to credit that, or it misjudges tasks like full parity.
- 3 A lower bound must hold for every trajectory**
Tracking the dynamics of a deep nonlinear transformer is out of reach. The proof has to sidestep them, through symmetry and gradient noise.

Each prior line answers a neighbouring question

Line of work	What it established	This paper's move
Expressivity: Merrill & Sabharwal; Feng et al.; Sanford et al. 2024	Transformers can represent connectivity at log depth, and P with long scratchpads	Ask whether training finds those weights
SQ and cross-predictability: Kearns 1998; Abbe & Sandon 2020	Noisy gradient descent fails on targets with low cross-predictability	A per-task measure for transformers that credits the token histogram
Leap and staircases: Abbe et al. 2022, 2023	SGD time for two-layer nets on isotropic data	A transformer analogue for token inputs and weak learning
Scratchpads and length generalization: Nye et al.; Anil et al.; Zhou et al.	Scratchpads help; extrapolation is fragile (parity 8 → 20 bits with a 128B model)	Say which scratchpads help, and give a format that extrapolates

One measure, one proved barrier, three kinds of scratchpad

GLOBALITY DEGREE

The fewest input tokens that, together with the token histogram, carry inverse-polynomial information about the label.

CONJECTURE 1

Weakly learnable by a regular transformer iff globality is constant.

THEOREM 1

Proved for a cycle-task variant: noisy gradient descent fails to weakly learn.

THEOREM 2

Unsupervised ("agnostic") scratchpad tokens leave the barrier in place.

SCRATCHPADS

Supervised steps of globality 2–3 learn; the inductive format also extrapolates.

Limitations, stated before the details

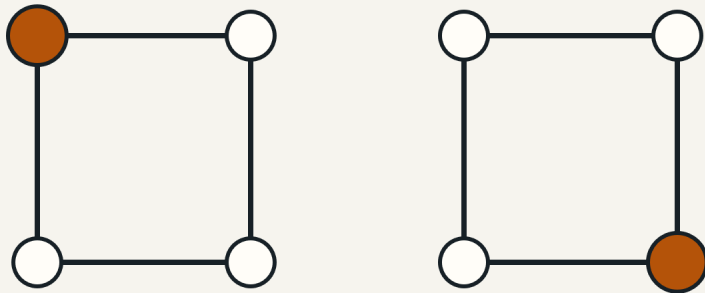
Scope	What the results assume
Model	"T-regular": polynomial size, i.i.d. Gaussian weights and positional embeddings, bidirectional attention, Lipschitz softmax
Training	Population gradient descent (or batches of n^c) with polynomial learning rate and clipping, plus gradient noise or random rounding
Task	Theorem 1 covers a purpose-built three-cycle variant, not the two-cycle task of Lemma 1 and the experiments
Claim	Weak learning only; the "iff" characterization is conjectured, and its positive direction has no proof
Inputs	Well-behaved inputs (no value frequent enough to memorize); constant alphabet for the positive direction
Experiments	Causal GPT-2-style decoders, outside the T-regular class; barrier plot only up to $n = 7$

What the paper contributes, and where each part is covered

Contribution	Result type	Slides
Setup: cycle task, T-regular transformers, scratchpad types	Preliminaries	11–15
Globality and autoregressive globality	Complexity measure	16, 27
Thm 1: barrier for the three-cycle variant	Learnability lower bound	17, 24–26
Conj. 1: learnable iff constant globality	Conjecture	18
Thm 2 and Conj. 2: agnostic scratchpads	Lower bound, conjecture	19, 28
Lemmas 2–3 and the inductive scratchpad	Construction, experiment	20, 28, 34
Syllogisms, cycle scaling, length generalization	Empirical	32–34
Our analysis: fine print, what carries the result, realism, adjacent work, one test	Critique	21–22, 29–40

Widom's guideline 1: the contribution is fully stated by slide 10 of 40. Technical depth follows.

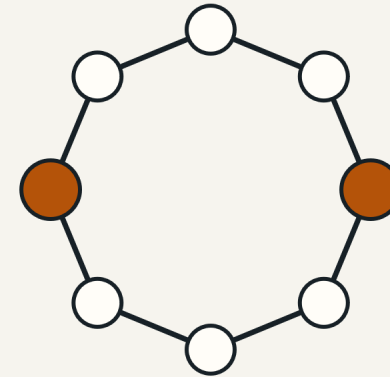
Running example: the cycle task with $n = 4$



Label 0: two 4-cycles; the query pair sits in different cycles

Why answers alone are hard

Vertex names are random. Any 3 of the 8 edges look the same in both classes; only all 4 edges of a cycle reveal the label.



Label 1: one 8-cycle; the query pair is 4 edges apart

What a scratchpad changes

Walk from one query vertex edge by edge. Each step reads the current vertex and the edge leaving it: 3 tokens.

The data-generating process

A sequence of distributions D_n on $A^n \times A$: n input tokens and one label token, alphabet size $|A| = \text{poly}(n)$.

Def. 1 · cycle task, size n

$2n$ vertices with random names; the input lists the edges and a query pair. Two equiprobable classes:

- Label 0: two disjoint n -cycles, query pair in different cycles
- Label 1: one $2n$ -cycle, query pair at distance n

Theorem 1's variant

One $3n$ -cycle with probability $2/3$, else three n -cycles. Query vertices a_0, b_0, c_0 (one per cycle, or pairwise n apart); vertices at distance i get random names a_i, b_i, c_i .

X = edges between consecutive layers; Y = whether a_0, b_0, c_0 share a cycle. Trivial accuracy $2/3$.

Also used: full and half parity, multi-digit addition, and random-graph syllogisms (24 edges on 24 vertices).

The hypothesis class: regular transformers and scratchpads

Def. 4 · T-regular

Polynomial size; weights and positional embeddings i.i.d. $N(0, 1)$; bidirectional attention; softmax Lipschitz in total variation.

With an s-scratchpad

Constant alphabet. Sample s tokens one at a time from the model's output distribution and append each; the last one is the answer.

Scratchpad	How its tokens are trained
Agnostic	Not supervised: the tokens only add memory and compute, trained through the final answer
Educated	Every token supervised, chosen so each step has low autoregressive globality
Inductive	Educated, plus $\langle \text{START} \rangle$ and $\#$ state tokens: each step sees only the question and the last state, with positions re-indexed

Loss, training rule and evaluation criterion

Element	Specification
Loss, Thm 1	Any differentiable loss
Rule, Thm 1	Population gradient descent, or batches of n^c ; polynomial learning rate and clipping; plus log-precision weights with random rounding, or polynomial gradient noise
Rule, Thm 2	SGD where, for each scratchpad position and each alternative symbol, the loss of that counterfactual scratchpad enters the update
Inputs	Well-behaved P_X (Def. 6): no input value frequent enough to weakly learn by memorizing it
Evaluation, theory	Weak learning: accuracy above the trivial accuracy by at least n^{-c} for some constant c
Evaluation, experiments	Iterations to $\geq 95\%$ accuracy with 512 fresh samples per iteration; accuracy on longer inputs than seen in training

A learnability result, with expressivity as the control

Category	What the paper contributes	Role
Representation	Nothing new; cites results that the cycle task is expressible	Control
Computational complexity	Links globality to NC^0 correlations; recalls TC^0 and TC^1 bounds	Framing
Trainability	Nothing	None
Optimization	Constrains noisy population gradient descent; no convergence result	Secondary
Learnability	Thms 1–2 (lower bounds); Conjs. 1–2 (characterization)	Primary
Generalization	Length generalization of the inductive scratchpad, measured only	Secondary
Certification	Nothing	None

Theorem 1 holds with exact population gradients: data is unlimited. The barrier is gradient noise or rounding, a precision barrier rather than a sample barrier.

Globality degree: how many tokens must be read together

$$\text{glob}(D) = \min \{ k : \exists S, |S| = k, I(X[S], P_X; Y) \geq n^{-0(1)} \}$$

P_X is the histogram of the input tokens. Autoregressive globality (Def. 3) applies the same test to each scratchpad step, over the input and earlier steps.

Task	Globality
Cycle task, size n	$\geq n$ (Lemma 1)
Full parity of n bits	1: the histogram counts the ones
Parity, running-parity scratchpad	2 per step (Lemma 2)
Cycle task, depth-first scratchpad	3 per step (Lemma 3)

Why the histogram: averaging attention computes token frequencies almost for free, so information available that way is not counted.

Why mutual information: any dependence counts, not only linear correlation; the threshold matches weak learning.

Theorem 1: the globality barrier, proved for one task

On the three-cycle variant of size n , **any** polynomial-size T -regular transformer, trained on **any** differentiable loss by population gradient descent (or batches of n^c) with polynomial learning rate and clipping, and with either log-precision weights under random rounding or polynomial gradient noise, **fails to weakly learn**: it cannot beat the trivial accuracy $2/3$ by n^{-c} .

Quantifiers

Universal over polynomial-size nets in the class, losses and polynomial hyperparameters.

Resources

Data is unlimited. Gradient noise or rounding is the binding constraint, as in SQ bounds.

Scope

One engineered task. The original cycle task has only Lemma 1 and experiments.

Conjecture 1: constant globality characterizes learnability

If P_X is well-behaved, then $P_{X,Y}$ is efficiently weakly learnable by a T-regular transformer **if and only if** $\text{glob}(P_{X,Y})$ is constant. The "if" direction assumes a constant alphabet.

"Only if": high globality $\Rightarrow$ hard

Proved for one task (Thm 1). Supported by the cycle-task scaling runs and by SQ intuition.

"If": constant globality $\Rightarrow$ learnable

No proof here. Later work proves cases in simplified one-layer models: parity with step-by-step CoT (Wen et al.; Kim & Suzuki).

Remark 3: the conjecture leans on permutation symmetry; causal masks or tailored positions may learn some higher-globality targets, such as parity of $\log n$ bits.

Agnostic scratchpads do not break the barrier

CONJECTURE 2

With an s-scratchpad trained by counterfactual-loss SGD and a well-behaved P_X , efficient weak learning still holds iff globality is constant.

Why it fails

The scratchpad is written by the same symmetric network, so its tokens carry no more information about the target than the output does.

THEOREM 2 (APP. F.1)

Proved for Theorem 1's task: the proof carries over almost unchanged, because the loss is an expectation over scratchpad values.

What it implies (interpretation)

Extra tokens trained only through the final answer, such as pause tokens or outcome-only latent reasoning, inherit the barrier on high-globality tasks.

Educated and inductive scratchpads keep every step local

EDUCATED · LEMMAS 2 AND 3

$y_i = y_{i-1} \cdot x_i \rightarrow \text{globality } 2$

Running parity: each step reads the previous partial result and one bit. Half-parity of 400 bits is learned this way.

DFS walk $\rightarrow \text{globality } 3$

Each step reads the current vertex and the edge leaving it. The cycle task is then learned as n grows.

INDUCTIVE · DEF. 5

$s[1] = g(Q, \emptyset), s[i] = g(Q, s[i-1])$

$Q \langle \text{START} \rangle s[1] \# s[2] \# \dots \# s[k] \langle \text{EOS} \rangle$

Masking shows each step only Q and the last state; positions restart at every state. The learned map g is blind to the step index, so more steps at test time just repeat it.

How Theorem 1 depends on each parameter

Parameter	Theorem 1 and Conjecture 1
Sample size	Unlimited: population gradients, or batches of n^c ; failure regardless
Steps	Polynomial, implied by "efficient"; no explicit rate
Size, depth, width	Any polynomial; the bound does not depend on them
Input length	$\Theta(n)$ edges; the cycle length n is also the globality lower bound
Precision	Log-precision weights with random rounding, or polynomial gradient noise
Learning rate, clipping	Polynomial
Alphabet	$\text{poly}(n)$ in Def. 2; constant for the conjecture's "if" direction
Conditioning	No condition number; i.i.d. Gaussian initialization
Task complexity	Globality: constant $\Rightarrow$ learnable (conjectured); $\geq n \Rightarrow$ not (Thm 1, one task)
Success threshold	Weak learning: margin n^{-c} over trivial accuracy

What the statements do not say

No rate

"Fails to weakly learn" with polynomial resources, but no explicit iteration bound: nothing to compare with a measured curve.

A different task

The theorem uses the layered three-cycle variant, built for its symmetry. The experiments use the two-cycle task.

A different learner

T-regular means bidirectional attention and random Gaussian positions. The experiments use causal GPT-2-style decoders.

A weak notion of success

Only weak learning is covered; strong learning (a "globality leap") is left as future work.

The proved barrier and the measured barrier concern different tasks and different learners. Only the conjecture links them.

Four results, three proof techniques

LEMMA 1 · COUNTING

Local indistinguishability

Any $n - 1$ edges have the same distribution in both classes, histogram included.

Slide 24

THEOREMS 1-2 · SYMMETRY

An orbit of targets noisy gradients cannot separate

Permutations preserve both data and initialization; their images of the target are nearly uncorrelated.

Slides 25–27

LEMMAS 2-3 · CONSTRUCTION

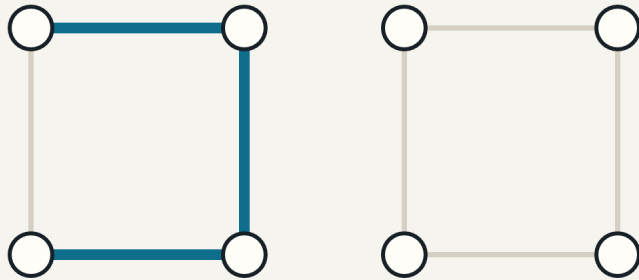
Steps that read three tokens

Exhibit a scratchpad whose every step is a function of at most three earlier tokens.

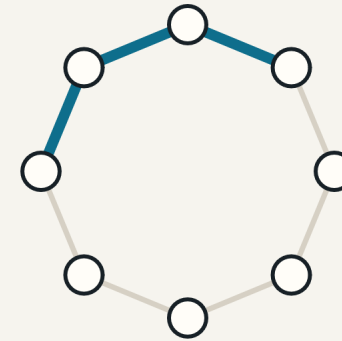
Slide 28

No proof tracks training dynamics. The lower bounds argue from what noisy gradients can reveal, as SQ bounds do.

Lemma 1: fewer than n edges look the same in both classes



Label 0: 3 edges of a 4-cycle form a path



Label 1: 3 edges of an 8-cycle form a path

Structure: fewer than n edges cannot close a cycle in either class, so they are paths, and random names make the two distributions coincide.

Histogram: every vertex appears in exactly two edges in both classes, so token counts carry nothing either.

So: $I(X[S], P_X; Y) = 0$ for every $|S| \leq n - 1$, hence $\text{glob} \geq n$.

Step 1: symmetries shared by the data and the network

The network

Positional embeddings are i.i.d. and attention is bidirectional, so permuting input positions leaves the initialization's distribution unchanged. Vertex names are random tokens, so renaming them is a symmetry too.

The data

In the three-cycle variant, swapping the non-fixed names between any two blocks leaves the input distribution unchanged. The two-cycle task lacks this exact symmetry, which is why the variant exists.

CONSEQUENCE

Training is equivariant: learning target $f \circ \pi$ is learning f with inputs relabeled by π . In effect the learner faces the whole orbit $\{f \circ \pi : \pi \in G\}$ at once.

Step 2: noisy gradients cannot tell orbit members apart

1

Low cross-predictability

For random π, π' in G , the targets $f \circ \pi$ and $f \circ \pi'$ are nearly uncorrelated: the label needs about n edges jointly, and relabeling scrambles which edges those are.

2

Gradients as statistical queries

Each update uses the population gradient $E[\nabla \ell]$. Averaged over the orbit, gradients for different targets differ negligibly; polynomial noise or random rounding hides the difference.

3

Same trajectory, so no learning

Training runs essentially the same whichever orbit member is the target, so the output cannot beat trivial accuracy by n^{-c} .

The histogram term is the symmetry argument in disguise

What symmetry leaves visible

A permutation-symmetric learner cannot tell which position a token came from, but averaging attention still reveals how often each token occurs. Information in the histogram is therefore free; only extra tokens read jointly cost anything.

What it predicts

Full parity has globality 1, since the count of ones decides it, so the conjecture predicts it is learnable. The cycle task keeps globality $\geq n$, because every vertex appears exactly twice in both classes.

Contrast with degree-based measures: full parity has maximal Fourier degree and sensitivity, yet minimal globality. Here the measures disagree.

Supervision, not extra tokens, is what lowers globality

AGNOSTIC

Each token is one more output the symmetric net must learn from the final loss alone. The orbit argument applies to the whole sequence, so nothing is gained.

EDUCATED

Each token gets its own target. Every step becomes a separate problem of globality ≤ 3 , learnable on its own if the conjecture's "if" direction holds.

INDUCTIVE

Same targets, plus masking and position resets, so every step is the identical problem whatever the chain length. The learned step transfers to longer chains.

Why the plain DFS scratchpad overfits: trained on 6-step walks, it sees absolute positions and the full history; at 12 steps those inputs are new. The inductive version only ever sees the question and the last state.

Symmetry and SQ-like training carry the barrier

Theorem 1 needs a learner that is blind to input positions **and** sees only noisy or rounded population gradients. Remove either and the proof no longer applies.

Permutation symmetry

i.i.d. Gaussian positional embeddings and bidirectional attention make every position interchangeable at initialization. This is what makes the orbit argument work (slide 25).

SQ-like training

Population gradients with polynomial noise or random rounding reveal only statistics of the data. That is what hides orbit members from each other (slide 26).

For the "if" direction

Well-behaved inputs and a constant alphabet keep low-globality targets from hiding hard functions in rare or huge token sets.

Every known escape breaks one of the assumptions

Escape	Assumption it breaks	Evidence
Supervised scratchpad	The data: steps get their own labels	DFS scratchpad learns the cycle task (Fig. 4a)
Masking and position resets	Symmetry: attention is restructured	Inductive scratchpad extrapolates (Figs. 4b, 5)
Mixing smaller instances	The data: lower-globality samples added	Weak learning within 30k iterations vs none in 100k (App. B.2)
Causal masks, tailored positions	Symmetry	Authors expect some higher-globality targets, e.g. parity of log n bits (Remark 3)
Small-batch SGD, special init	SQ-like training	Can emulate any poly-time learner, parities included (Abbe & Sandon 2020)

The paper's own fixes sit outside the T-regular class: the inductive scratchpad changes attention and positions, which Theorem 1 assumes fixed.

How closely a T-regular transformer matches an LLM

Component	T-regular (theory)	Modern LLM	Load on the result
Positions	i.i.d. Gaussian, random	learned or rotary	High: symmetry drives the proof
Attention	bidirectional	causal	High: order becomes visible
Start point	Gaussian, from scratch	pretrained	High: globality uses raw tokens
Training	noisy population descent	minibatch Adam, bf16	High: SQ-likeness assumed
Size	any polynomial	billions of parameters	Low: the bound is size-free
Goal	weak learning, one task	high accuracy, many tasks	Medium

Plausibly transfers: answer-only training on tasks that need many tokens at once invites shortcuts; supervised local steps help.

Not established: any guarantee for causal, pretrained models, where raw-token globality may be the wrong coordinates.

What the barrier experiment can and cannot show

$n \leq 7$

largest cycle size in the scaling plot: six problem sizes, $n = 2$ to 7

10–85M

parameters across three causal GPT-2-style models, all outside the T-regular class

5.1×10^7

samples at the 100k-iteration cap, with 512 fresh samples per iteration

Six sizes cannot separate super-polynomial growth from a steep polynomial, and the theorem states no rate to compare against.

What the plot shows: difficulty rising fast with n for practical models. What it cannot show: an asymptotic barrier.

Barrier experiments: what was measured and what it shows

Experiment	Result	Relation to the theory
Random-graph syllogisms, 24 edges (App. B.1)	About 80% accuracy; a degree heuristic alone scores about 82% and tracks the model; chance out of distribution	Shortcut learning; motivates the cycle task
Cycle task, $n = 2-7$ (Fig. 1b)	Iterations to 95% rise fast; the 10M model makes no progress at $n = 7$ within 100k	Consistent with Lemma 1 and Conj. 1; outside Thm 1's class
DFS scratchpad (Fig. 4a)	The cycle task is learned as n grows	Lemma 3: steps of globality 3
Mixed sizes 2–7 (App. B.2)	Weak learning within 30k iterations; size 7 alone showed none in 100k	Lower-globality samples help

Widom's test: the runs vary n and model size. None uses a T-regular model or varies gradient noise and batch size, the settings Theorem 1 depends on.

Length generalization: strong numbers, task-specific formats

EXPERIMENT · TRAINED → TESTED LENGTH

Setting	Length
Parity, NoPE (Kazemnejad et al.)	8 → 12 bits
Parity, 128B model + scratchpad (Anil et al.)	8 → 20 bits
Parity, RASP-L + index hints (Zhou et al.)	30 → 50 bits
Parity, inductive + random spaces	30 → 50–55 bits
Addition, inductive + random spaces	10 → 18 digits
Addition, inductive + shifts	4 → 26 digits

Shaded rows: this paper (Tables 1–2, Fig. 5)

OUR CRITIQUE

Formats carry part of the gain. Random spaces and cyclic shifts are hand-designed layouts; the authors note that settings and model sizes differ across rows.

The state is hand-designed. An inductive scratchpad presupposes the recurrence g: more annotation than an educated one.

Seeds matter. 55-bit parity and 20-digit addition hold only for some seeds.

Proved, conjectured, measured, or interpreted

PROVED

- Thm 1: barrier for the three-cycle variant
- Thm 2: agnostic scratchpads, same task
- Lemma 1: cycle globality $\geq n$
- Lemmas 2–3: scratchpads of globality 2 and 3

CONJECTURED

- Conj. 1: learnable iff constant globality
- Conj. 2: the same with agnostic scratchpads
- The "if" direction in general

MEASURED

- Degree shortcuts on syllogisms
- Fast-rising cost, cycle task $n \leq 7$
- DFS scratchpad and size mixtures help
- $30 \rightarrow 55$ -bit parity; $4 \rightarrow 26$ -digit addition

INTERPRETED

- "Transformers cannot compose long syllogism chains"
- Inductive scratchpads as the right bias for reasoning
- Globality as the explanation for CoT in LLMs

The title's question is answered for regular transformers trained from scratch; the leap to LLMs sits in the last column.

Relation to adjacent work

Paper	Relation	How
Merrill & Sabharwal; Sanford et al.	Contrasts	The cycle task is representable, yet has globality $\geq n$
Abbe & Sandon 2020	Changes assumptions	Same lower-bound engine, driven by transformer symmetry
Abbe et al. 2022–23 (leap)	Advances	Transformer analogue: tokens plus histogram, not isotropic inputs
Hahn & Rofin 2024	Contrasts	Sensitivity calls full parity hard; globality calls it easy
Zhou et al. (RASP-L)	Contrasts	RASP-L excludes full parity; globality predicts it learnable
Nye et al.; Wei et al.	Reframes	Scratchpads typed by supervision; only local, supervised steps help
Wen et al.; Kim & Suzuki	Later special cases	Prove the "if" direction for step-wise parity CoT, in simplified models

Full parity is where the measures disagree most, which makes it the cleanest test between them.

Three measures of hardness, three architectures

Axis	Globality (this paper)	Leap (Abbe et al. 2023)	Sensitivity (Hahn & Rofin)
Counts	tokens read jointly, beyond the histogram	new coordinates per SGD stage	input bits whose flip changes the output
Model	T-regular transformers	two-layer nets, isotropic data	transformers
Predicts	weak learning in polynomial time	SGD time $\approx d^{\max(\text{Leap}, 2)}$	sharp minima, poor generalization
Full parity	easy: globality 1	hardest: leap n	hardest: sensitivity n
Status	conjecture plus one theorem	conjecture, proved for a class	theorems on the landscape

Each measure encodes an architecture's invariances. Averaging attention makes full parity cheap for a symmetric transformer; nothing does for an MLP on isotropic data.

CONCLUSION

What is now known, and what is not

PROVED

On a purpose-built cycle task, no polynomial-size regular transformer trained by noisy population descent beats trivial accuracy by n^{-c} . The plain cycle task needs $\geq n$ tokens jointly; supervised scratchpads with 2–3-token steps exist.

OBSERVED

A 10M causal model made no progress on 7-cycles in 100k iterations; with supervised DFS steps it learns. Inductive formatting extends parity from 30 to 55 bits and addition from 4 to 26 digits.

OPEN

The conjecture's "if" direction, and whether the barrier survives causal masks, learned positions, small-batch training and pretraining.

POINT 8 · ONE PRECISE LIMITATION

Theorem 1 binds only learners that are blind to position and see only noisy population gradients. Every experiment, and the paper's own fix, uses causal decoders with structured positions.

So the theorem covers a learner no one trains in practice, and the experiments cover practical learners only up to $n = 7$. Conjecture 1 is the only bridge between them.

POINT 8 · ONE CONCRETE EXPERIMENT

A two-factor ablation that tests the barrier on real learners

Cycle task, $n = 2$ to 12 , fresh samples. Cross (A) bidirectional with frozen random positions vs (B) causal with learned or rotary positions, against (i) large-batch descent with injected gradient noise vs (ii) small-batch Adam. Three model sizes, five seeds; record iterations to trivial + 5% and to 95%; fit log-iterations against n and against $\log n$.

ALL FOUR CELLS BLOW UP · STRENGTHENS

The barrier holds beyond its proof. Next theorem: extend Theorem 1 to causal attention and small-batch SGD.

ONLY (A)(i) BLOWS UP · CHALLENGES

The barrier is an artifact of symmetry and noise. Globality would need redefining relative to the architecture's invariances.

Control: full parity, which the conjecture predicts is learnable in every cell.